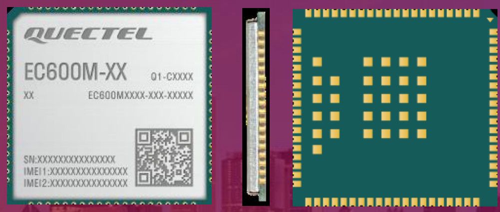
Quectel EC600M Series

IoT/ M2M-optimized

LTE Cat 1 bis Module



EC600M-CN-LE_QuecPython Release Notes

LTE Standard Module Series

Rev. EC600M-CN-LE_QuecPython_Firmware_Release_Notes_V0604

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Our aim is to provide customers with timely and comprehensive service. For any assistance, please contact our company headquarters:

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Or our local office. For more information, please visit:
http://www.quectel.com/support/sales.htm.

For technical support, or to report documentation errors, please visit:
http://www.quectel.com/support/technical.htm
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Contents

1. Release Content

This document provides the Release Notes for EC600M-CN-LE_QuecPython. The current release includes the following firmware packages.

Package	Firmware Version
Firmware	EC600MCNLER06A04M08_OCPU_QPY
Toolchain	Helios_toolchain_1.4.1.exe

2. Matters Needing Attention

SN	Item
[1]	To prevent the frequently erase and write of SIM card due to the frequent changes in network information, affecting the service life of the SIM card, the network information is not updated by default, and GUTI and other network information are saved when the module is turned off normally with *Power.powerDown()*. The normal shutdown method is recommended.
[2]	To extend the service life of flash, it is recommended that the total number of operations related to power-on/off, CFUN switching, SIM card hot plugging, or dual-SIM switching should not exceed 30 times per day.
[3]	Please make sure that the APN has been set correctly before network registration.
[4]	There are two upgrade packages for upgrading via mini-system.
[5]	For UART transmission, it is recommended to add hardware flow control to reduce the risk of data loss during transmission.
[6]	The module turns on the remap function by default. This function is suitable for most operators' LTE network scenarios.
[7]	During the FOTA upgrade process, it is necessary to ensure the stable power supply of the module. If the power is disconnected during the upgrade, there is a small probability that the flash will be damaged.
[8]	During the module shut-down process, the main UART output garbled data when it is turned off.
[9]	The 32 KHz (clock source) output from the module has an error within 20 PPM. Do not provide it to the peripherals with high accuracy demand.

3. Release History

3.1. Firmware Release History

Firmware Version	Description
EC600MCNLER06A04M08_OCPU_QPY	Mass production
EC600MCNLER06A03M08_OCPU_QPY	Mass production
EC600MCNLER06A02M08_OCPU_QPY	Mass production

3.2. New Features

EC600MCNLER06A04M08_OCPU_QPY	
Item	**Brief Description**
FOTA	Add *ipvtype* to the constructors of the request and app_fota modules. The default value of *ipvtype* is 0, that is, IPv4, and *ipvtype* can be set to 1, that is, IPv6.
HTTP	Support IPv6 in the request module.
I2C	Add the I2C3 interface, with SLC connected to Pin57 and SDA connected to Pin58. The existing I2C1 interface has SLC connected to Pin75 and SDA to Pin74.
SSL	Add the feature of CA certificate configuration in the ussl module.
General	Add *utime.localtime_ex()* and *utime.mktime_ex()* to solve the problem that the old functions may cause time conversion errors in scenarios of non-UTC+8 zones because they are only applicable to the UTC+8 zone.
General	Add the 32 KHz clock signal output feature.
EC600MCNLER06A03M08_OCPU_QPY	
Item	**Brief Description**
Network	Add *nic.status()* to query the status of the W5500 Ethernet NIC.
General	Add the half-hour timekeeping functions *utime.setTimeZoneEx()* and *utime.getTimeZoneEx()*.
EC600MCNLER06A02M08_OCPU_QPY	
Item	**Brief Description**
/	/

3.3. Improved Features

EC600MCNLER06A04M08_OCPU_QPY	
Item	**Brief Description**
Low Power	Add *misc.Power.camVDD2V8Enable()* to set the voltage output from the cam_vdd interface
PING	Limit the range of *SIZE* for the PING feature.

EC600MCNLER06A03M08_OCPU_QPY	
Item	**Brief Description**
MiniFOTA	Solve the problem of firmware download failure caused by network registration using blank APN in specific areas through setting the user-configured APN information before the MiniFOTA network is established.
Network	Solve the problem of USBnet function failure.
(U)SIM	Solve the problem that the operator name could not be returned after the execution of *net.operatorName()* or **AT+COPS?** when a SIM card of China Broadcast Network Co., Ltd was inserted.
General	Solve the problem that one extra row and one extra column were configured when the matrix keyboard was initialized.

EC600MCNLER06A02M08_OCPU_QPY	
Item	**Brief Description**
/	/

3.4. Known Issues

Item	Brief Description
/	/

NOTE

Verification Environment is shown below. For more details, please contact Quectel technical support.
USB Driver: Quectel_Windows_USB_DriverA_Customer_V1.1.13
QPYcom Tool: QPYcom_V3.6

4. Functions List

Category	Item	Supported Version (Since)	Note
Basic Function	(U)SIM	EC600MCNLER06A02M08_OCPU_QPY	/
	Data Call	EC600MCNLER06A02M08_OCPU_QPY	/
	FILE	EC600MCNLER06A02M08_OCPU_QPY	/
Protocol Function	ALiYun	EC600MCNLER06A02M08_OCPU_QPY	/
	HTTP	EC600MCNLER06A02M08_OCPU_QPY	/
	HTTPS	EC600MCNLER06A02M08_OCPU_QPY	/
	MQTT	EC600MCNLER06A02M08_OCPU_QPY	/
	NITZ	EC600MCNLER06A02M08_OCPU_QPY	/
	NTP	EC600MCNLER06A02M08_OCPU_QPY	/
	PING	EC600MCNLER06A02M08_OCPU_QPY	/
	SSL	EC600MCNLER06A02M08_OCPU_QPY	/
	TCP/UDP	EC600MCNLER06A02M08_OCPU_QPY	/
	Tencent Cloud	EC600MCNLER06A02M08_OCPU_QPY	/
Special Function	ADC	EC600MCNLER06A02M08_OCPU_QPY	/
	Alarm	EC600MCNLER06A02M08_OCPU_QPY	/
	Audio Playback	EC600MCNLER06A02M08_OCPU_QPY	/
	Audio Record	EC600MCNLER06A02M08_OCPU_QPY	/
	Camera	EC600MCNLER06A02M08_OCPU_QPY	/
	Device Info	EC600MCNLER06A02M08_OCPU_QPY	/
	EINT	EC600MCNLER06A02M08_OCPU_QPY	/
	FOTA	EC600MCNLER06A02M08_OCPU_QPY	/
	GNSS	EC600MCNLER06A02M08_OCPU_QPY	/
	GPIO	EC600MCNLER06A02M08_OCPU_QPY	/

	I2C	EC600MCNLER06A02M08_OCPU_QPY	/
	Low Power	EC600MCNLER06A02M08_OCPU_QPY	/
	MiniFOTA	EC600MCNLER06A02M08_OCPU_QPY	/
	Modem	EC600MCNLER06A02M08_OCPU_QPY	/
	PWM	EC600MCNLER06A02M08_OCPU_QPY	/
	SIM Detection	EC600MCNLER06A02M08_OCPU_QPY	/
	SPI	EC600MCNLER06A02M08_OCPU_QPY	/
	TTS	EC600MCNLER06A02M08_OCPU_QPY	/
	Timer	EC600MCNLER06A02M08_OCPU_QPY	/
	UART	EC600MCNLER06A02M08_OCPU_QPY	/
	USB	EC600MCNLER06A02M08_OCPU_QPY	/
	Wi-Fi Scan	EC600MCNLER06A02M08_OCPU_QPY	/
	Encryption and Decryption	EC600MCNLER06A02M08_OCPU_QPY	/

About Quectel

Quectel's passion for a smarter world drives us to accelerate IoT innovation. As a highly customer-centric organization, we are a global IoT solution provider backed by outstanding support and services. Our growing global team of 5,900 professionals sets the pace for innovation in cellular, automotive, smart, GNSS, satellite, Wi-Fi and Bluetooth modules as well as antennas and services.

With regional offices and support across the globe, our international leadership is devoted to advancing IoT and building a smarter world.